MAP DATA LIBRARY

Database MAP_DATA_WELD

1. Provenance of Data

Tracey Cool* and H. K. D. H. Bhadeshia, Phase Transformations Group, Department of Materials Science and Metallurgy, University of Cambridge, Cambridge, U.K.

*TC is now with the Materials Engineering Department, Parsons Power Generation Systems Ltd, Heaton Works, Shields Road, Newcastle Upon Tyne, NE6 2YL

2. Purpose

Provides chemical composition and mechanical property data for a collection of all weld metal deposits.

3. Description

The tar file contains data for the chemical composition of the steels studied, and their room temperature mechanical properties. These data have been collated from available literature. The presence of an "N" indicates that the value was not reported in the publication. This is *not* meant to be an indication that the value is zero. Although many columns are presented here, those which only have a small amaount of data have been removed for the analysis described in [1 and 2]. This enabled more input lines to be used for neural network analysis.

Column 1

Carbon concentration / weight %

Column 2

Silicon concentration / weight %

Column 3

Manganese concentration / weight %

Column 4

Sulphur concentration / weight %

Column 5

Phosphorus concentration / weight %

Column 6

Nickel concentration / weight %

Column 7

Chromium concentration / weight %

Column 8

Molybdenum concentration / weight %1

Column 9

Vanadium concentration / weight %

Column 10

Copper concentration / weight %

Column 11

Cobalt concentration / weight %

Column 12

Tungsten concentration / weight %

Column 13

Oxygen concentration / parts per million by weight

Column 14

Titanium concentration / parts per million by weight

Column 15

Nitrogen concentration / parts per million by weight

Column 16

Aluminium concentration / parts per million by weight

Column 17

Boron concentration / parts per million by weight

Column 18

Niobium concentration / parts per million by weight

Column 19

Tin concentration / parts per million by weight

Column 20

Arsenic concentration / parts per million by weight

Column 21

Antimony concentration / parts per million by weight

Column 22

Current / A

Column 23

Voltage / V

Column 24

AC or DC

Column 25

Electrode positive or negative

Column 26

Heat input / kJ mm^{-1}

Column 27

Interpass temperature / °C

Column 28

Type of weld; ShMA = MMA = manual metal arc, SA = SMA = submerged arc, FCA = flux cored arc, GTAA gas tungsten arc automatic, GMAA gas metal arc automatic, SAW-NG = submerged arc narrow gap, GMA-NG = gas metal arc narrow gap, ES = electroslag, TSA = tandem submerged arc.

Column 29

Post weld heat treatment temperature / °C

Column 30

Post weld heat treatment time / hours

Column 31

Yield strength / MPa

Column 32

Ultimate tensile strength / MPa

Column 33

Elongation / %

Column 34

Reduction of Area / %

Column 35

Charpy temperature / °C

Column 36

Charpy impace toughness / J

Column 37

Hardness / kg mm^{-2}

Column 38

50 % FATT

Column 39

Primary ferrite in microstructure / %

Column 40

Ferrite with second phase / %

Column 41

Acicular ferrite / %

Column 42

Martensite / %

Column 43

Ferrite with carbide aggreagate / %

Column 44

Weld ID

4. References

1. Tracey Cool, Design of Steel Weld Deposits, PhD Thesis, University of Cambridge, 1996

2. Tracey Cool, H. K. D. H. Bhadeshia and David J. C. MacKay, Materials Science and Engineering, 1997

5. Further Comments

None.

6. Keywords

all weld metal deposits, yield strength, UTS, elongation, charpy toughness